

# **The evolution of polyandry in animals: When is it beneficial, and why is it rare?**

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## **Abstract**

Polyandry is an important model for understanding the evolution of mating systems because it challenges conventional sex-role expectations and reveals how ecological, social, and genetic benefits can reshape reproductive strategies. Previous research has identified multiple benefits of polyandry, but no single explanation accounts for both its repeated evolution and rarity across lineages. This dissertation explores the evolution of polyandry across different taxa, to compare and discuss different drivers, benefits and costs associated with this mating strategy. Research points to polyandry evolving through three main routes: social/cooperative polyandry, female desertion with partial or full sex-role reversal, and genetic or fertility-related polyandry. Benefits of social polyandry include increased male parental care, cooperative behaviour and reduced intrasexual aggression while costs can be paternity confusion and reduced litter size. Female desertion is associated with permissive ecological conditions such as food abundance, mate availability and shifts in OSR that make females gain through sequential breeding possible. Lastly, genetic polyandry, where a female mates with genetically different males, can be beneficial by providing fertility assurance, increasing genetic compatibility and avoidance of infertile males while in some cases such as predation-induced male-biased OSR, it may take the form of convenience polyandry in socially monogamous birds. The examples in this dissertation show a unified evolutionary explanation for the rarity of polyandry: the benefits only outweigh the costs under narrow, unstable or taxon specific conditions.

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## **1) INTRODUCTION**

Polyandry is a mating system in which females mate with multiple males within a breeding period (Taylor, Price and Wedell, 2014; Slatyer et al., 2012). Classical sexual selection theory predicts that anisogamy, a pattern in which females produce larger, costlier gametes while males produce smaller, more numerous gametes, should favour greater female investment and greater male competition (Darwin, 1871; Bateman, 1948; Trivers, 1974).

Darwin introduced sexual selection as variation in reproductive success arising from mate competition and choice while Bateman argued that male reproductive success often increases more strongly with number of mates. Afterwards, Trivers linked sex roles to parental investment and more recently, Kokko and Jennions argued that anisogamy alone cannot explain all variation in sex roles (Darwin, 1871; Bateman, 1948; Trivers, 1974; Kokko and Jennions, 2008). Polyandry is therefore important because it challenges a widely accepted model in which females are always choosy and males always compete and so the central puzzle of this dissertation is formed: if it provides reproductive benefits, why is it not more widespread, and if it is costly, why does it evolve at all?

To explore this dilemma, the relationship between concepts such as operational sex ratio (OSR) and anisogamy must be understood. OSR refers to the ratio of sexually receptive males to sexually receptive females, which shapes mating competition (Trivers, 1974; Emlen and Oring, 1977). Anisogamy usually limits female egg production and increases maternal investment while OSR often contains more available males than females; thus, polyandry is not expected to be favourable. However, a pattern regularly seen in polyandrous systems is female competition and limitation of male reproductive success which lead to departures from conventional sex roles (Trivers, 1974; Emlen and Oring, 1977; Goymann et al., 2015; Fresneau et al., 2021). Therefore, polyandry shouldn't be considered as an exception outside sexual selection theory, but a result of the factors that shape sex roles being shifted (Kokko and Jennions, 2008).

This dissertation will endeavour to explain why polyandry has evolved repeatedly across taxa yet remains rare in most lineages using selected case studies. It approaches this question through three routes: social benefits, ecological conditions linked to sex-role reversal or female desertion, and genetic or fertility-related benefits (Garber, 1997; Emlen and Oring, 1977; Slatyer et al., 2012). These routes are compared using selected case studies, including callitrichids and eclectus parrots for social polyandry, coucals, pipefish and jacanas for sex-role reversal, and ground crickets, flour beetles and swift parrots for genetic and fertility pathways (Garber, 1997; Heinsohn et al., 2019; Fedorka and Mousseau, 2002; Vasudeva et al., 2021; Goymann, 2021). This structure allows polyandry to be treated not as one uniform mating system, but as a multifaceted evolutionary mechanism. The following sections therefore examine how each route can make multiple mating advantageous before considering why those conditions are not more widespread.

**Table 1. Comparative summary of polyandrous mating systems and their proposed fitness consequences across selected animal taxa.** Table 1 compares social/cooperative, sex-role-reversed or female-desertion-associated, and genetic/fertility-related forms of polyandry. For each taxon, the main driver is summarized alongside the male role, potential female benefit and principal cost or limitation. Created in Biorender.com.

| Type of Polyandry                         | Taxa                            | Main Driver                                                     | Male Role                                | Female Benefit                                               | Main Cost                                      |
|-------------------------------------------|---------------------------------|-----------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------|------------------------------------------------|
| Social & Cooperative                      | Tamarins & Marmosets            | High infant-care costs, twin births, short interbirth intervals | Infant carrying, food sharing, vigilance | Higher offspring survival through shared care                | Female-female competition                      |
|                                           | Eclectus Parrots                | Scarce, variable nest hollows                                   | Provision females at nest hollows        | Attraction of multiple males & production of more fledglings | Depends on rare high-quality nesting sites     |
| Female Desertion and/or Sex-role-reversal | Snowy Plovers & Tengmalm's Owls | Early breeding or food abundance                                | Male care after female desertion         | Additional breeding attempts                                 | Current brood may suffer reduced survival      |
|                                           | Black Coucals                   | Food abundance, high nest loss                                  | Male-only incubation and chick care      | More clutches & rapid replacement after nest failure         | Requires surplus males and male care ability   |
|                                           | Wide-bodied pipefish            | Limited brood-pouch capacity                                    | Males brood embryos & court females      | Increased reproductive success                               | Sexual conflict or harassment                  |
| Genetic and/or Fertility Benefits         | Ground Crickets                 | Genetic compatibility & nuptial gifts                           | Sperm contribution & nuptial gifts       | Higher offspring survivorship                                | Female longevity cost; sexual conflict         |
|                                           | Swift Parrots                   | Male-biased ASR as a result of predation                        | Extra-pair males seek mating access      | No clear benefit                                             | Sexual conflict & reduced population viability |
|                                           | Flour Beetles                   | Heatwave-induced male subfertility                              | Sperm donor without parental care        | Fertility assurance                                          | Benefit is facultative                         |

## 2) MECHANISMS OF SOCIAL AND COOPERATIVE POLYANDRY

Social and cooperative polyandry can evolve when mating with multiple males provides direct benefits such as infant care, provisioning or group stability. In these systems, polyandry is not only a mating strategy, but also a way of organising parental effort around demanding reproductive conditions. Tamarins and eclectus parrots illustrate two different routes: one driven by high infant-care demands, and the other by scarce breeding resources.

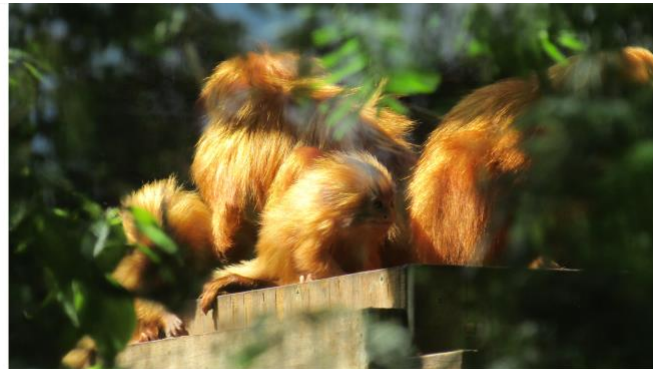
### 2.1. Tamarins: high infant-care demands

Polyandry in tamarins is driven by the need for multiple carers under high infant-care demands. The dominant breeding female typically produces twins and may have two litters per year, creating a care burden intensified by infant transport over 1.5–2 km per day as groups move through the forest (Goldizen, 1990; Garber, 1997; Garber et al., 2016; **Table 1**). During the first month, twins are often carried together before being carried singly for shorter periods, making multiple adult helpers valuable for offspring survival (Garber et al., 2016; **Table 1**).

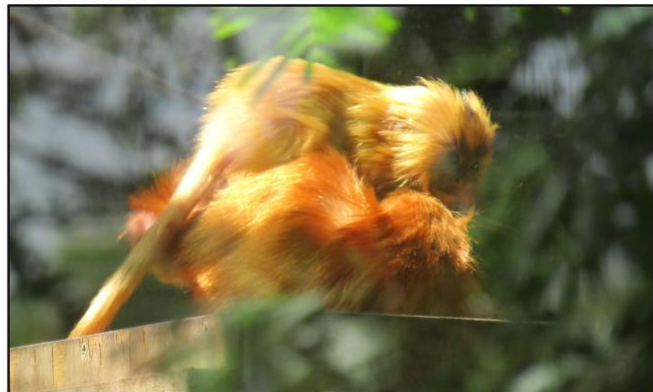
Uncertain paternity through polyandry can promote male care: while males may also compete through care if investment increases their future reproductive opportunities (Garber, 1997; Garber et al., 2016), polyandry may also give reproductive access to males that might otherwise gain none. Through these mechanisms, tamarin polyandry functions as a “care-concentration” system, where one breeding female receives support from multiple alloparents. Although polygyny or multiple breeding females could increase infant production, this might reduce offspring survival if care becomes diluted. Polyandry can therefore improve offspring survival and reduce female-female competition (**Table 1**).



**Figure 1A:** Infants from the reported 2026 triplet litter clinging to an adult during movement, illustrating the carrying burden associated with callitrichid reproduction.



**Figure 1B:** Two adults and two infants resting in close proximity, illustrating the social tolerance and group context that can support cooperative care in callitrichids.



**Figure 1C:** An infant being carried by an adult, demonstrating the infant-care demands that can make alloparental support beneficial in golden lion tamarins.

**Figure 1. Social context and infant transport in golden lion tamarins at London Zoo.**

During observation, the group was reported by staff to have produced triplets in 2026, providing a contemporary example of how larger litters can intensify care demands. Photographs by the author, London Zoo, 12 May 2026; figure assembled using BioRender.com.

Cooperative polyandry in reverse-dichromatic eclectus parrots is instead driven by resource scarcity. The females depend on rare tree hollows for breeding, and hollow quality affects reproductive success because wetter hollows are more likely to flood, reducing egg or chick survival (Heinsohn, Legge and Endler, 2005; **Table 1**). Females occupying drier, higher-quality hollows therefore control valuable breeding resources and attract multiple provisioning males. Females benefit from increased food supply and defence, while males may gain partial paternity by provisioning a high-value female when they might otherwise gain none (Heinsohn, Legge and Endler, 2005).

Unlike tamarins, where polyandry is linked to infant-care demand, eclectus parrots show that social polyandry can also emerge when males gain limited reproductive access through provisioning. However, this route remains rare because it depends on extreme nest-site limitation and a situation where males gain more from sharing access than from competing or abandoning the female.

### **2.3. Reduced intrasexual aggression**

Social polyandry also relies on reduced intrasexual aggression. If male-male conflict is intense, it can disrupt care, reduce group stability and prevent multiple males from investing in the same offspring (Garber et al., 2016). In moustached tamarins, Garber observed only 19 cases of within-group aggression from almost 8,000 activity records in *Saguinus mystax*, with aggression forming just 0.2% of social interactions. By contrast, cooperative behaviours such as food sharing, group defence and infant transport made up 11.5% of activities, giving a cooperative-to-aggressive act ratio of 52:1 (Garber, 1997; Figure 1). This indicates that reproductive sharing may be stabilised by cooperation, and that male care can be more valuable than exclusive mating attempts in these groups. This contrasts with swift parrots, where genetic polyandry occurs alongside male conflict rather than cooperative care, showing that the social context of polyandry strongly affects its fitness consequences.

Together, tamarins and eclectus parrots demonstrate that social polyandry can evolve through different direct benefits, but both depend on rare conditions: multiple males must gain enough from shared reproduction to cooperate and continue providing care or resources.

### **3) Drivers of female desertion and sex-role reversal**

Female desertion occurs when a female leaves the current brood in the care of the male to pursue other reproductive opportunities. When resources are abundant, females may shift from investing only in the current brood to sequential breeding because food reduces the energetic cost of egg production and allows males to feed themselves while caring for offspring (Andersson, 2005; Eldegard and Sonerud, 2009; Goymann et al., 2015). This is supported by many polyandrous birds, including jacanas and black coucals, occupying food-rich habitats (Goymann et al., 2015; Fresneau et al., 2021; **Table 1**). However, food abundance alone cannot explain polyandry, because sequential female breeding also requires male availability for remating as well as other ecological factors. Food abundance should therefore be treated as a permissive condition rather than a sufficient cause.

#### **3.1. The evolution of female desertion**

When these conditions align, polyandry can evolve through female desertion if the benefit of producing another brood exceeds that of continued care (Eldegard and Sonerud, 2009; Eberhart-Hertel et al., 2023). This may also help explain smaller eggs in some polyandrous birds, such as snowy plovers, because producing smaller eggs can shorten laying intervals and allow females to complete clutches more quickly (Andersson, 2004; Goymann et al., 2015; Eberhart-Hertel et al., 2023). In snowy plovers, desertion is also strongly linked to timing: early breeding females lay smaller eggs and are more likely to pursue sequential breeding. (Eberhart-Hertel et al., 2023). Tengmalm's owls provide a supporting example: when food is abundant and breeding conditions allow, females may leave males and attempt further reproduction (Eldegard and Sonerud, 2009). These examples introduce the logic seen more strongly in black coucals, where male-only care, food abundance, nest predation and seasonality interact to produce sex-role-reversed polyandry. Desertion is therefore not automatically adaptive, and it should also be distinguished from sex-role reversal: females may desert opportunistically without becoming consistently more competitive or causing males to become strongly choosy.

#### **3.2. The relationship between sex-role reversal and polyandry**

Sex-role reversal occurs when male reproductive capacity becomes more limiting than female reproductive capacity (Trivers, 1974). In wide-bodied pipefish, females are ornamented while males brood embryos through "male pregnancy". Polyandry becomes beneficial when females produce eggs faster than males can brood, biasing the operational sex ratio towards available

females and creating competition for non-brooding males who may become choosier (van Eyndhoven, Cameron and Flanagan, 2023). However, sex-role reversal should be treated as a spectrum rather than a complete inversion of all behaviours as male pipefish initiated 61% of reciprocated courtship bouts, showing that male courtship can persist in a sex-role-reversed system (van Eyndhoven, Cameron and Flanagan, 2023). Similarly, for pheasant-tailed jacanas that have an average of 2.4 mates per female and at least 81.8% of females being polyandrous, males still spent significantly more time on courtship behaviours than females (Fresneau et al., 2021). Male-only care results in a female-biased OSR, where the ratio can influence the intensity of sex-role reversal through competition. This pattern highlights that paternal care strong enough to make males reproductively limiting under a specific alignment of care capacity, OSR and ecological conditions can lead to the evolution of sex-role-reversed polyandry.

### **3.3 Black coucals vs white-browed coucals: a perfect example?**

The comparison between black coucals and mostly monogamous white-browed coucals shows why this alignment matters (Goymann et al., 2015; Goymann, 2021). In sex-role-reversed black coucals, seasonal food abundance allows females to leave one male incubating and seek another mate, making them more likely to be preyed upon due to being the more mobile and defensive sex (Goymann et al., 2015; **Table 1**). Although this shifts adult sex ratio towards males, incubation can make sexually available males limiting, producing the female-biased OSR associated with sex-role reversal. Nesting ecology is also crucial: black coucals experienced 49% nest loss, compared with 32% in white-browed coucals, possibly because white-browed coucals nest in thorny acacia shrubs rather than more exposed high grass (Goymann et al., 2015). These pressures interact with a short breeding season and high breeding density, creating conditions where females multiple mating is advantageous. The contrast is important: food abundance has been proposed as a key factor in avian polyandry (Emlen and Oring, 1977; Erckmann, 1983; Jenni, 1974), however black and white-browed coucals show that it is not sufficient on its own, since only black coucals show sex-role-reversed polyandry despite both species occupying food-rich habitats.

Overall, snowy plovers, Tengmalm's owls, pipefish, jacanas and coucals illustrate the argument that female desertion and sex-role-reversed polyandry are shaped by ecological and life-history conditions that relieve the constraints that often make female multiple mating unfavourable.

## **4) Genetic and fertility-related routes to polyandry**

Evidence for genetic benefits of polyandry are mixed. Slatyer et al. (2012) found that genetic benefits of polyandry can be variable and often small across experimental studies, suggesting that indirect benefits alone may not consistently favour female multiple mating. Taylor, Price and Wedell, (2014) reported an association between polyandry and multilocus heterozygosity across major animal groups, but this does not prove that polyandry directly causes increased genetic quality. Holman et al. (2015) modelled whether polyandry could reduce the spread of X-linked meiotic drive but found that polyandry alone was insufficient. Genetic benefits are therefore possible, but not universally strong or sufficient, potentially requiring additional selective pressure to cause a direct effect. Nonetheless, in precarious systems like polyandry, indirect genetic benefits may be the grain that tips the scale and need to be studied in more species to allow better comparison.

### **4.1. Material benefits, nuptial gifts and polyandry in ground crickets**

The example of ground crickets illustrates clearly how genetic polyandry can improve reproductive success (Fedorka and Mousseau, 2002; **Table 1**). Females chew the male's tibial spur, a nuptial gift, during copulation and gain direct material benefits (Fedorka and Mousseau, 2002). Females mated to multiple males showed a 2.4-fold increase in hatching success, possibly due to genetic compatibility, sperm competition, or avoidance of potentially infertile males. Because hatching success was independent of nuptial gift size, the benefit appears to come from mating with genetically different males rather than from increased material benefits (**Table 1**). However, polyandrous females had shorter longevity than females repeatedly mated to one male, reinforcing the pattern that polyandry can carry direct costs that need to be overcome or reduced by the benefits.

#### **4.2. Facultative polyandry: a fertility assurance mechanism**

Flour beetles show that polyandry can evolve facultatively as fertility assurance under heat stress. Vasudeva et al. (2021) exposed males to heatwave conditions and compared females given access to either one or five males. Heat exposure damages male fertility, making monandry risky if the single male is subfertile; however, females given access to five heat-damaged males restored offspring production to normal levels compared with females mated to one heat-damaged male (Vasudeva et al., 2021; **Table 1**). Females also increased remating after mating with heatwave-exposed males, although the cue remains uncertain. This indicates flexible fertility assurance rather than fixed polyandry. However, storing sperm from multiple males did not protect females if stored sperm was later damaged by heat inside the reproductive tract, so the benefit came from new mating opportunities after male fertility was compromised. Polyandry was beneficial under male subfertility, but not when males were normal and fertile, helping explain why polyandry can be context dependent and possibly an adaptive mechanism.

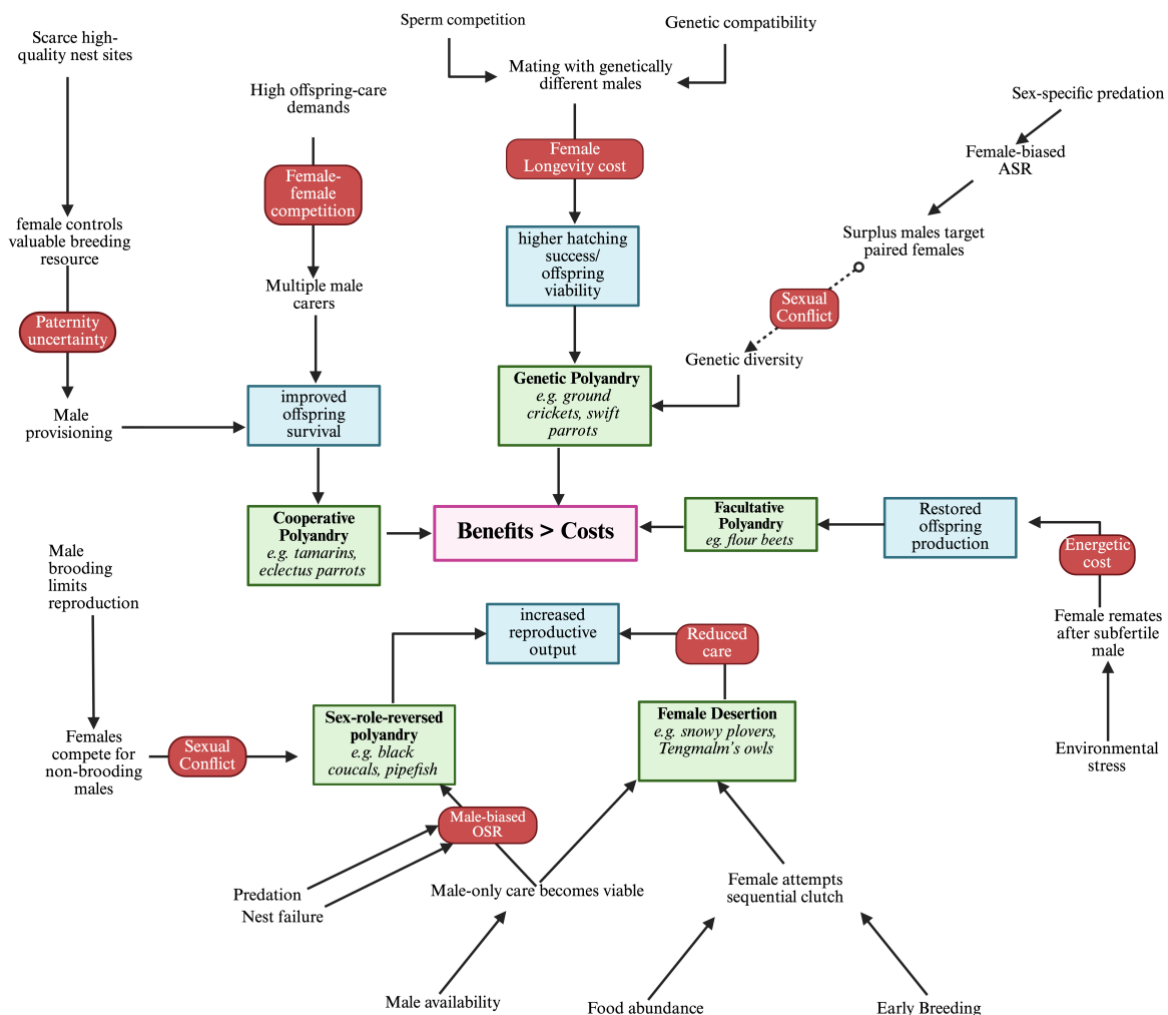
#### **4.3. Swift parrots: polyandry without benefits?**

Genetically polyandrous but socially monogamous swift parrots appear to be pushed towards polyandry by a predation-driven, male-biased adult sex ratio (ASR) (Heinsohn et al., 2019). Because nesting females are disproportionately preyed upon, regions with greater predation show higher shared paternity, suggesting that surplus males increasingly target already-paired females. Heinsohn et al. (2019) found shared paternity in 50.5% of nests and estimated the ASR to be at least 73% male (**Table 1**). Higher shared paternity was associated with worse projected population declines: models predicted an 89.4% decline over three generations at the lowest observed shared paternity rate, compared with 92.1–94.9% under higher shared paternity rates. Even unpredated nests produced fewer fledglings in areas with more male-biased sex ratios. This suggests that introduced predation can indirectly reshape mating systems by altering survival, sex ratio, and mating opportunities.

Unlike eclectus parrots, swift parrots remain socially monogamous: resident males chase unpaired males from nest areas, while females may accept food and mate with interlopers. This resembles convenience polyandry, where rejecting persistent males may be costlier than

accepting mating, rather than cooperative polyandry based on clear female choice or benefit (Table 1). However, causation should be interpreted cautiously because pre-introduction shared paternity is unknown, and reduced fledging success may be driven by the same predation pressure and male-biased ASR that increase polyandry, rather than by polyandry itself. Swift parrots therefore show that genetic polyandry can arise through demographic stress without benefits, even if potential genetic-diversity advantages cannot be fully excluded.

Overall, genetic benefits may contribute to polyandry when multiple mating improves offspring viability, hatching success, genetic diversity or fertility assurance, but these benefits are often conditional, indirect or accompanied by costs.



**Figure 2. Conditional pathways through which polyandry may evolve.** Green boxes show forms of polyandry, blue boxes show potential benefits or outcomes, and red boxes show costs or constraints. Solid arrows indicate proposed pathways, dashed arrows show uncertain links. Figure assembled using BioRender.com.

## 5) Conclusion

As seen in figure 2, polyandry is best understood as a set of context-dependent mating strategies rather than a single system with one evolutionary cause. In the case studies examined here, ground crickets and flour beetles illustrate genetic or fertility-related routes, while tamarins, eclectus parrots and coucals show how social care, resource limitation and ecology can favour social or sex-role-reversed polyandry. This fits the wider pattern that polyandry is repeatable because different taxa can reach it through different mechanisms, but no single mechanism is broadly sufficient.

This dissertation is limited by the fact that it compares and analyses polyandry across markedly different animal groups with varying levels of social complexity, parental care, and taxon-specific behaviour, making direct comparisons between case studies necessarily cautious. In some cases, causation is also difficult to deduce as effectors of polyandry are interlinked. Future research should aim to compare how interactions between drivers influence the evolution of the type of polyandry observed as well as exploring whether its benefits differ according to species complexity and social organization.

The case studies analyzed therefore support a conditional and flexible view of polyandry: it can evolve when scarce breeding resources, skewed OSRs and ASRs or demographic disruption make benefits such as provisioning, paternal care, sequential breeding or fertility assurance outweigh reproductive and survival costs. Thus, polyandry remains rare because the conditions that make female multiple mating beneficial are narrow, unstable or taxon specific.

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